JOSEF DOORNINK

Site Reliability Engineer | AI Infrastructure & MLOps

Portland, OR | jdoorarg@gmail.com | LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

A creative problem-solver who loves building the infrastructure that brings ideas into reality. An engineer with 10+ years experience, 10 peer-reviewed papers, 2 US patents, a major design award and FDA-cleared hardware. That foundation now drives deep SRE and AI infrastructure work (CKS/CKA certified), delivering systems with integrity and observability at scale - currently designing autonomous Agentic SRE pipelines that leverage LLMs for root-cause analysis.

TECHNICAL SKILLS

Core Competencies: Site Reliability Engineering | Large-Scale Distributed Systems | AI Platform Engineering | Performance Optimization | Profiling & Instrumentation | Systems Engineering | Team Productivity Tools

Languages: Python | Go/Golang | Bash | PowerShell | C# | SQL

GPU Infrastructure & LLM Serving: vLLM | GPU Sizing & Capacity Planning | KV Cache Tuning | Model Quantization (AWQ) | KEDA / HPA Autoscaling | Inference Load Testing | GPU Node Pools (AKS)

ML/AI Engineering: LLM Model Serving | ML Pipeline Development | Model Training Infrastructure | Azure Machine Learning | Distributed Training Systems | Model Finetuning Systems

Distributed Systems & Orchestration: Kubernetes (AKS, Production-Scale) | Docker | Helm | Kafka | Redis | SQL Databases | Service Mesh

Cloud & Infrastructure: Azure (Expert) | AWS | Terraform (IaC) | Infrastructure as Code | Configuration as Code | ARM Templates

DevOps & SRE: GitHub Actions | Azure DevOps | CI/CD Pipelines | Monitoring (Prometheus, New Relic, Azure Monitor) | Logging | Alerting | Incident Response

Systems & Tools: Linux | Git | kubectl | Nginx | Performance Profiling | Distributed Tracing | Load Testing

PROFESSIONAL EXPERIENCE - STARTUP

Lead MLOps Engineer

Reason Benefit AI Corporation | Remote | October 2025 - February 2026

- Architect and maintain large-scale Azure Kubernetes Service (AKS) production environment for ML model training and serving, supporting distributed model inference at scale
- Build Python-based automation tools for ML pipeline orchestration, reducing manual overhead by 70% and accelerating model deployment velocity
- Integrate with observability frameworks for model performance tracking, latency monitoring, and resource utilization across distributed training systems
- Collaborate with research teams to translate experimental model architectures into production-ready systems with focus on reliability and scalability
- Implement automated testing frameworks for ML pipelines to quickly detect regressions and ensure model quality in production

PROFESSIONAL EXPERIENCE

Lead Site Reliability Engineer (SRE) I -> II -> III
Trimble | Portland, OR | January 2019 - Present

- Architected and maintained large-scale Azure Kubernetes Service (AKS) production environments handling 10M+ requests/day across 30+ microservices with 99.9% uptime SLA
- Developed high-performance automation tools using **Python** and **Go** that eliminated 80+ hours/month of operational toil, accelerating deployment velocity by 3x across engineering teams
- Led performance optimization initiatives through systematic profiling and instrumentation, reducing P99 latency by 45% and improving throughput by 60% for distributed systems
- Built custom CLI tooling in **Go (Cobra framework)** to streamline workflows for 50+ engineers, dramatically improving team productivity through better developer experience
- Designed and implemented comprehensive observability stack (New Relic) with distributed tracing for debugging performance bottlenecks in distributed microservices
- Led capacity planning and performance optimization for Kubernetes clusters and backend databases, implementing auto-scaling strategies supporting 200% traffic growth
- Created sophisticated CI/CD pipelines using **GitHub Actions** and Azure DevOps with automated testing, sophisticated deployment templates, and rollback mechanisms
- Implemented Infrastructure as Code using **Terraform** managing 500+ cloud resources, enabling consistent and repeatable infrastructure provisioning
- Participated in on-call rotation providing critical incident response, root cause analysis, and post-mortem documentation for production systems
- Mentored software developers on distributed systems reliability, SRE principles, and performance optimization techniques
- Championed pair programming culture and led technical design reviews for architecture proposals

Software Developer
Viewpoint | Portland, OR | March 2018 - January 2019

- Developed cloud-based SaaS applications using .NET and Angular, migrating on-premise software solutions to Azure cloud platform
- Built RESTful APIs for multi-tenant applications serving thousands of users with focus on performance and scalability
- Implemented front-end features using Angular integrated with .NET backend services
- Collaborated with cross-functional teams to deliver high-quality software solutions on time

Software Developer I
Onfulfillment | Portland, OR | March 2014 - March 2018

- Engineered multi-tenant e-commerce platform using Microsoft Stack (.NET, C#, SQL Server) integrated with third-party SaaS APIs
- Led "uplift" initiative migrating legacy codebase to modern greenfield platform, improving response times by 40% measured through New Relic APM
- Developed automated testing frameworks and implemented performance monitoring to identify and resolve bottlenecks
- Created project plans for large-scale refactoring initiatives and communicated technical progress to stakeholders

KEY TECHNICAL PROJECTS

vLLM on Kubernetes: GPU Sizing & Autoscaling (2026)
Repository | Article on Dev.to

- Served a quantized Qwen2.5-7B-Instruct-AWQ model on vLLM across A10 GPU node pools in AKS, published as an engineering article series with a reproducible deployment
- Derived a first-principles GPU sizing method - weight memory vs. KV cache headroom converted to concurrent-request capacity - so GPU selection is a calculation rather than a guess
- Eliminated autoscaler flapping by replacing lagging queue-depth triggers with a leading KV-cache-utilization signal: KEDA scaling on sum(vllm:kv_cache_usage_perc) at 0.80 with 120s scale-down damping, whose fixed point is invariant to replica count
- Built an async load generator and Grafana instrumentation to characterize inference saturation and validate stable vs. flapping scaling behavior

Distributed Training Pipeline Optimization (2025)
Reason Benefit AI Corporation

- Reduced training time by 35% through systematic profiling and optimization of RL pipeline bottlenecks
- Decreased Python runtime contention by refactoring GIL-constrained components with multiprocessing techniques
- Improved delivery confidence by building automated validation for training pipeline health

Kubernetes Security Hardening Initiative (2024)
Trimble

- Reduced security vulnerabilities by 85% through systematic Kubernetes hardening and continuous monitoring
- Applied CKS best practices across production clusters to support SOC2 compliance requirements
- Improved response speed by automating security scanning and remediation workflows

High-Performance CLI Tooling (2023)
Trimble

- Increased engineering productivity by delivering production-grade Go CLI tools for infrastructure operations
- Added service discovery, log aggregation, and deployment automation capabilities used in daily workflows
- Drove adoption across 50+ engineers and improved operational efficiency through better developer UX

CERTS/COURSES

- **CNCF Certified Kubernetes Security Specialist (CKS)** | March 2024 | LF-ghgugl1a0s
- **CNCF Certified Kubernetes Administrator (CKA)** | June 2021 | LF-w50bpv1lpd
- **Machine Learning Specialization** | Stanford/Coursera | September 2025
- **Microsoft Certified Azure Developer Associate** | August 2019 | H210-5692
- **HashiCorp Certified Terraform Associate** | July 2022 | HCTAO-002
- **Production Machine Learning Systems** | Google Cloud / Coursera | April 2026 | 01X41BH2KFVE

EDUCATION

Master of Science, University of California, Davis | 2006

Bachelor of Science, Mechanical Engineering | California State University, Chico | 2003

PUBLICATIONS

- M Bottlang, **J Doornink**, DC Fitzpatrick, SM Madey. *Far cortical locking can reduce stiffness of locked plating constructs while retaining construct strength.* The Journal of Bone and Joint Surgery (JBJS), 2009
- M Bottlang, **J Doornink**, GD Byrd, DC Fitzpatrick, SM Madey. *A nonlocking end screw can decrease fracture risk caused by locked plating in the osteoporotic diaphysis.* The Journal of Bone and Joint Surgery (JBJS), 2009
- DC Fitzpatrick, **J Doornink**, SM Madey, M Bottlang. *Relative stability of conventional and locked plating fixation in a model of the osteoporotic femoral diaphysis.* Clinical Biomechanics, 2009
- M Bottlang, M Lesser, J Koerber, **J Doornink**, B von Rechenberg, P Augat, DC Fitzpatrick, SM Madey, JL Marsh. *Far cortical locking can improve healing of fractures stabilized with locking plates.* The Journal of Bone and Joint Surgery (JBJS), 2010
- M Bottlang, **J Doornink**, TJ Lujan, DC Fitzpatrick, JL Marsh, P Augat, B von Rechenberg, M Lesser, SM Madey. *Effects of construct stiffness on healing of fractures stabilized with locking plates.* The Journal of Bone and Joint Surgery (JBJS), 2010
- **J Doornink**, DC Fitzpatrick, S Boldhaus, SM Madey, M Bottlang. *Effects of hybrid plating with locked and nonlocked screws on the strength of locked plating constructs in the osteoporotic diaphysis.* Journal of Trauma and Acute Care Surgery, 2010
- **J Doornink**, DC Fitzpatrick, SM Madey, M Bottlang. *Far cortical locking enables flexible fixation with periarticular locking plates.* Journal of Orthopaedic Trauma, 2011
- PJ Denard, **J Doornink**, D Phelan, SM Madey, DC Fitzpatrick, M Bottlang. *Biplanar fixation of a locking plate in the diaphysis improves construct strength.* Clinical Biomechanics, 2011
- PA Wackym, JA Ratigan, JD Birck, SH Johnson, **J Doornink**, M Bottlang, SK Gardiner, FO Black. *Rapid cVEMP and oVEMP responses elicited by a novel head striker and recording device.* Otology & Neurotology, 2012
- M Bottlang, DC Fitzpatrick, D Sheerin, E Kubiak, R Gellman, C Vande Zandschulp, **J Doornink**, K Earley, SM Madey. *Dynamic fixation of distal femur fractures using far cortical locking screws: a prospective observational study.* Journal of Orthopaedic Trauma, 2014

PATENTS & AWARDS

Patents:

- *Bone Screw with Multiple Thread Profiles for Far Cortical Locking and Flexible Engagement to a Bone.* US Patent No. US10918430B2 (2021), US9314286B2 (2016), US8740955B2 (2014)

Awards:

- *Scientific Exhibit Award of Excellence*, American Academy of Orthopaedic Surgeons (AAOS), 2010

FDA-Approved Medical Devices:

- Led engineering team that created 4 FDA-approved orthopaedic implants currently used in national and international trauma centers (K101696, K123918, K130810)

ADDITIONAL EXPERIENCE

Biomechanical Research Engineer II | Legacy Biomechanics Research Lab | Portland, OR | 2007-2013

- Lead test and development engineer for NIH-funded multimillion-dollar research project focused on bone fixation solutions
- Managed successful implant creation, delivery, and test methodology producing multiple US FDA-approved implants
- Developed automated testing protocols and data collection systems using MTS software

Honorary Fellow | BG Unfallklinik | Murnau, Germany | 2008

- Established mechanical analysis protocols for orthopaedic research measuring torsional strength and stiffness of bone specimens
- Documented test protocols for process automation and results collection

Quality Assurance Associate | Google | Mountain View, CA | 2006-2007

- Evaluated accuracy of Google search engine results and web layout effectiveness
- Collaborated remotely with interdisciplinary web development teams

REFERENCES: Available upon request